

SAFETY DATA SHEET



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DATE PREPARED: 7/15/2020
SDS No: AFCO 9031

Section 1 – Chemical Product and Company Information

PRODUCT NUMBER: AFCO 9031
PRODUCT NAME: AFCOOL Shield
CATEGORY: Cooling Water Treatment Product

COMPANY NAME: AFCO | Zep
COMPANY ADDRESS: 800 Development Avenue
COMPANY CITY: Chambersburg, PA 17201
COMPANY PHONE: (800) 345-1329

EMERGENCY PHONE: CHEMTREC (US) (800) 424-9300

Section 2 – Hazards Identification

Physical Hazards:

Flammable liquids Category 4

Health Hazards:

Skin corrosion/irritation Category 1

Serious eye damage/eye irritation Category 1

Label elements:



Signal Word:

DANGER

Hazard Statement(s):

Combustible liquid. Causes severe skin burns and eye damage. Causes serious eye damage.

Precautionary Statement(s):

Prevention

Keep away from flames and hot surfaces-No smoking. Do not breathe mist or vapor. Wash thoroughly after handling. Wear protective gloves/protective clothing/eye protection/face protection.

Response

If swallowed: Rinse mouth. Do NOT induce vomiting. If on skin (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. If inhaled: Remove person to fresh air and keep comfortable for breathing. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a poison center/doctor. Wash contaminated clothing before reuse. In case of fire: Use appropriate media to extinguish.

Storage

Store in a well-ventilated place. Keep cool. Store locked up.

Disposal

Dispose of contents/container in accordance with local/regional/national/international regulations.

Hazard(s) not otherwise classified (HNOC)

None known.

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Supplemental information

None.

Section 3 - Composition/Information on Ingredients

Mixtures

CAS #	Chemical name:	Common name and synonyms	%
37971-36-1	2-Phosphono-1,2,4-butanetricarboxylic acid		2.5 - 10
61791-26-2	Polyoxyethylene tallow amine		2.5 - 10
61791-63-7	N-coco-1,3-diaminopropane		1 - 5
68424-85-1	Quaternary ammonium compounds, benzyl-C12-16-alkyldimethyl, chlorides		1 - 5
Other components below reportable levels			80 - 100

*Designates that a specific chemical identity and/or percentage of composition has been withheld as a trade secret.

Section 4 – First Aid Measures

Emergency and First Aid Procedures:

Inhalation:

Move to fresh air. Call a physician if symptoms develop or persist.

Skin Contact:

Take off immediately all contaminated clothing. Rinse skin with water/shower. Call a physician or poison control center immediately. Chemical burns must be treated by a physician. Wash contaminated clothing before reuse.

Eye Contact:

Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Call a physician or poison control center immediately.

Ingestion:

Call a physician or poison control center immediately. Rinse mouth. Do not induce vomiting. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs.

Indication of immediate medical attention and special treatment needed:

Provide general supportive measures and treat symptomatically. Chemical burns: Flush with water immediately. While flushing, remove clothes which do not adhere to affected area. Call an ambulance. Continue flushing during transport to hospital. Keep victim under observation. Symptoms may be delayed.

General information:

Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

Section 5 – Fire Fighting Measures

Suitable extinguishing media:

Water fog. Foam. Dry chemical powder. Carbon dioxide (CO2).

Unsuitable extinguishing media:

Do not use water jet as an extinguisher, as this will spread the fire.

Specific hazards arising from the chemical:

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The product is combustible, and heating may generate vapors which may form explosive vapor/air mixtures. During fire, gases hazardous to health may be formed.

Special protective equipment and precautions for firefighters:

Self-contained breathing apparatus and full protective clothing must be worn in case of fire.

Fire fighting equipment/instructions:

In case of fire and/or explosion do not breathe fumes. Move containers from fire area if you can do so without risk.

Specific methods:

Use standard firefighting procedures and consider the hazards of other involved materials.

General fire hazards:

Combustible liquid.

Section 6 – Accidental Release Measures

Personal precautions, protective equipment and emergency procedures:

Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.

Methods and materials for containment and cleaning up:

Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. This product is miscible in water.

Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Use a non-combustible material like vermiculite, sand or earth to soak up the product and place into a container for later disposal. Following product recovery, flush area with water.

Small Spills: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination.

Never return spills to original containers for re-use. For waste disposal, see section 13 of the SDS.

Environmental precautions:

Avoid discharge into drains, water courses or onto the ground.

Section 7 – Handling and Storage

Precautions for safe handling:

Keep away from open flames, hot surfaces and sources of ignition. When using do not smoke. Do not get in eyes, on skin, or on clothing. Avoid prolonged exposure. Provide adequate ventilation. Wear appropriate personal protective equipment. Observe good industrial hygiene practices.

Conditions for safe storage, including any incompatibilities:

Store locked up. Keep away from heat, sparks and open flame. Store in a cool, dry place out of direct sunlight. Store in tightly closed container. Store in a well-ventilated place. Keep in an area equipped with sprinklers. Store away from incompatible materials (see Section 10 of the SDS).

Section 8 – Exposure Controls/Personal Protection

Occupational exposure limits:

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The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. Workplace Environmental Exposure Level (WEEL) Guides:

Components	Type	Value	From
2-Phosphono-1,2,4-butanet ricarboxylic acid (CAS 37971-36-1)	TWA	10 mg/m3	Aerosol.

Biological limit values:

No biological exposure limits noted for the ingredient(s).

Appropriate engineering controls:

Good general ventilation should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Eye wash facilities and emergency shower must be available when handling this product.

Individual protection measures, such as personal protective equipment

Eye/face protection:

Wear safety glasses with side shields (or goggles) and a face shield.

Skin protection

Hand protection:

Wear appropriate chemical resistant gloves.

Other:

Wear appropriate chemical resistant clothing.

Respiratory protection:

If engineering controls do not maintain airborne concentrations below recommended exposure limits (where applicable) or to an acceptable level (in countries where exposure limits have not been established), an approved respirator must be worn.

General hygiene considerations:

When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

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Section 9 – Physical and Chemical Properties

Physical state	Liquid.
Form	Clear Liquid.
Color	Yellow.
Odor	Characteristic
Odor threshold	Not available.
pH	2.0 - 3.0
Melting point/freezing point	< 40 °F (< 4.4 °C) estimated
Initial boiling point and boiling range	> 212 °F (> 100 °C) estimated
Flash point	> 179.6 °F (> 82 °C) Cleveland Closed Cup
Evaporation rate	Not available.
Flammability (solid, gas)	Not applicable.
Upper/lower flammability or explosive limits	
Flammability limit – lower (%)	Not available.
Flammability limit – upper (%)	Not available.
Explosive limit - lower (%)	Not available.
Explosive limit - upper (%)	Not available.
Vapor pressure	< 1.0 mm Hg estimated
Vapor density	Not available.
Relative density	Not available.
Solubility(ies) Solubility (water)	Complete
Partition coefficient (n-octanol/water)	Not available.
Auto-ignition temperature	Not available.
Decomposition temperature	Not available.
Viscosity	Not available.
Other information	
Density	8.47 lb/gal
Explosive properties	Not explosive.
Flammability class	Combustible IIIA estimated
Oxidizing properties	Not oxidizing.
Percent volatile	> 80 % estimated
Specific gravity	1.01

Section 10 – Stability and Reactivity

Reactivity:

The product is stable and non-reactive under normal conditions of use, storage and transport.

Chemical stability:

Material is stable under normal conditions.

Possibility of hazardous reactions:

No dangerous reaction known under conditions of normal use.

Conditions to avoid:

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Avoid heat, sparks, open flames and other ignition sources. Avoid temperatures exceeding the flash point. Contact with incompatible materials.

Incompatible materials:

Strong oxidizing agents.

Hazardous decomposition products:

No hazardous decomposition products are known.

Section 11 – Toxicological Information

Information on likely routes of exposure

Inhalation:

May cause irritation to the respiratory system. Prolonged inhalation may be harmful.

Skin contact:

Causes severe skin burns.

Eye contact

Causes serious eye damage.

Ingestion:

Causes digestive tract burns.

Symptoms related to the physical, chemical and toxicological characteristics:

Burning pain and severe corrosive skin damage. Causes serious eye damage. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Permanent eye damage including blindness could result.

Information on toxicological effects

Acute toxicity:

Not available.

Skin corrosion/irritation:

Causes severe skin burns and eye damage.

Serious eye damage/eye irritation:

Causes serious eye damage.

Respiratory or skin sensitization

Respiratory sensitization:

Not a respiratory sensitizer.

Skin sensitization:

This product is not expected to cause skin sensitization.

Germ cell mutagenicity:

No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.

Carcinogenicity:

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Not classifiable as to carcinogenicity to humans.

IARC Monographs. Overall Evaluation of Carcinogenicity:

Not listed.

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1052)

Not regulated.

US. National Toxicology Program (NTP) Report on Carcinogens

Not listed.

Reproductive toxicity:

This product is not expected to cause reproductive or developmental effects.

Specific target organ toxicity - single exposure:

Not classified.

Specific target organ toxicity - repeated exposure:

Not classified.

Aspiration hazard:

Not an aspiration hazard.

Chronic effects:

Prolonged inhalation may be harmful.

Section 12 – Ecological Information

ALL WORK PRACTICES MUST BE AIMED AT ELIMINATING ENVIRONMENTAL CONTAMINATION

Ecotoxicity:

The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.

Persistence and degradability:

No data is available on the degradability of any ingredients in the mixture.

Bioaccumulative potential Mobility in soil:

No data available.

Other adverse effects:

No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

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Section 13 – Disposal Considerations

Disposal instructions:

Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Dispose of contents/container in accordance with local/regional/national/international regulations.

Local disposal regulations:

Dispose in accordance with all applicable regulations.

Hazardous waste code:

The waste code should be assigned in discussion between the user, the producer and the waste disposal company.

Waste from residues / unused products:

Dispose of in accordance with local regulations. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner (see: Disposal instructions).

Contaminated packaging:

Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

Section 14 – Transport Information

DOT

Not regulated as dangerous goods.

IATA

Not regulated as dangerous goods.

IMDG

Not regulated as dangerous goods.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not established.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not regulated as dangerous goods.

Section 15 – Regulatory Information

US federal regulations:

This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.

CERCLA Hazardous Substance List (40 CFR 302.4):

Not listed.

SARA 304 Emergency release notification:

Not regulated.

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OSHA Specifically Regulated Substances (29 CFR 1910.1001-1052):

Not regulated.

Superfund Amendments and Reauthorization Act of 1986 (SARA)

SARA 302 Extremely hazardous substance:

Not listed.

SARA 311/312 Hazardous chemical:

Yes

Classified hazard categories:

Flammable (gases, aerosols, liquids, or solids)
Skin corrosion or irritation
Serious eye damage or eye irritation

SARA 313 (TRI reporting):

Not regulated.

Other federal regulations

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List

Not regulated.

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130):

Not regulated.

Safe Drinking Water Act (SDWA):

Contains component(s) regulated under the Safe Drinking Water Act.

US state regulations

US. California Controlled Substances. CA Department of Justice (California Health and Safety Code Section 11100)

Not listed.

California Proposition 65:

California Safe Drinking Water and Toxic Enforcement Act of 2016 (Proposition 65): This material is not known to contain any chemicals currently listed as carcinogens or reproductive toxins.

International Inventories

Country(s) or region:

United States & Puerto Rico

Inventory name:

Toxic Substances Control Act (TSCA) Inventory

On inventory (yes/no)*:

Yes

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*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s) A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

Section 16 – Other Information

Issue date:
07/06/2020

Version #:
01

Disclaimer:

AFCO | Zep cannot anticipate all conditions under which this information and its product, or the products of other manufacturers in combination with its product, may be used. It is the user's responsibility to ensure safe conditions for handling, storage and disposal of the product, and to assume liability for loss, injury, damage or expense due to improper use. The information in the sheet was written based on the best knowledge and experience currently available.